COMPUTER NETWORKING

Course Project Report

Network Monitoring and Performance Analysis

Tool Used: Cisco Packet Tracer

Environment: Simulated Network

1. Introduction

Network monitoring and performance analysis is a critical aspect of modern computer networking. It involves observing network traffic, measuring performance metrics, identifying bottlenecks, and detecting faults to ensure optimal network availability and efficiency.

1.1 Objectives

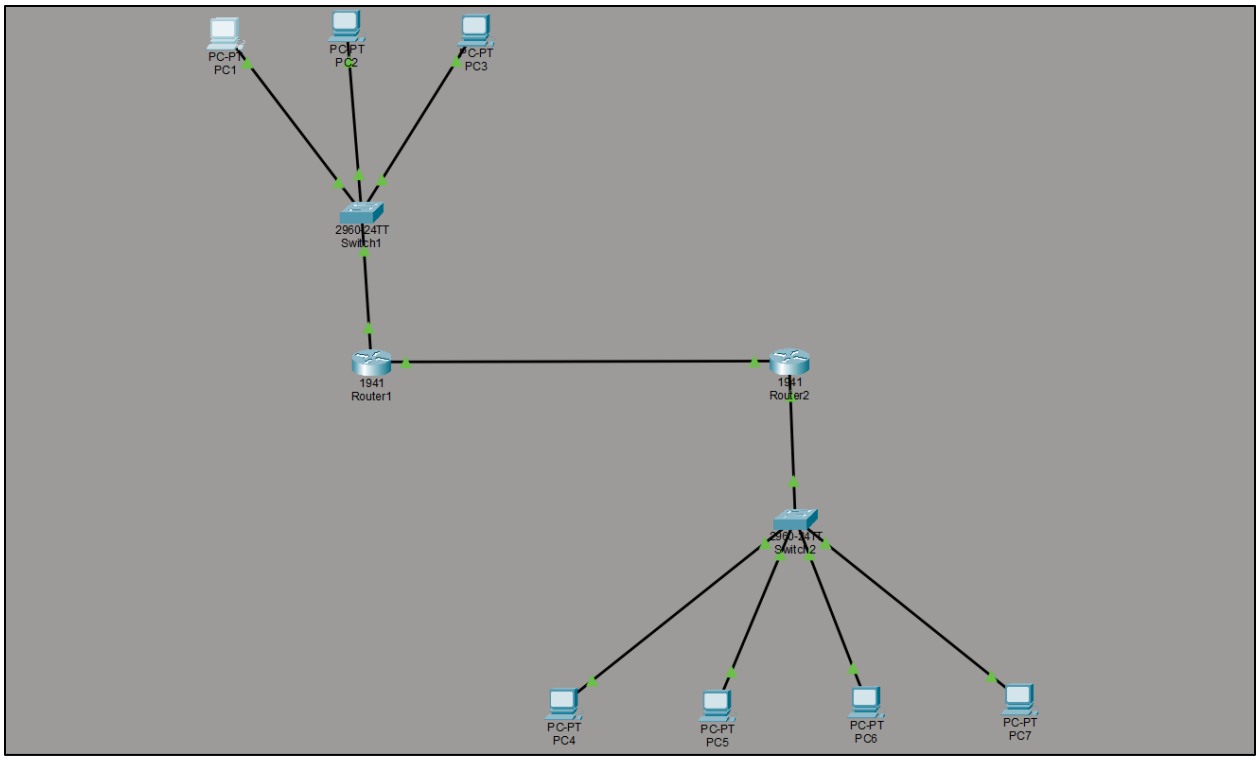
- Monitor network traffic using Cisco Packet Tracer Simulation Mode
- Analyze performance metrics including RTT, TTL, and hop count
- Identify network bottlenecks using Traceroute
- Detect and resolve network faults
- Ensure optimal network availability and efficiency

1.2 Tool Used

Cisco Packet Tracer was used as the simulated network environment for this project. It provides a realistic platform to build, configure, and monitor networks without requiring physical hardware.

2. Network Topology

The following network topology was designed and built in Cisco Packet Tracer for this project:



2.1 Devices Used

Device	Quantity	Purpose
Router (1941)	2	Routing between networks
Switch (2960)	2	LAN switching
PC	7	End user devices

2.2 Cable Connections

From	To	Cable Type
PC1, PC2, PC3	Switch1	Copper Straight-Through
Switch1	Router1 (Gig0/0)	Copper Straight-Through
Router1 (Gig0/1)	Router2 (Gig0/0)	Copper Straight-Through
Router2 (Gig0/1)	Switch2	Copper Straight-Through
Switch2	PC4, PC5, PC6, PC7	Copper Straight-Through

3. IP Address Configuration

3.1 Network 1 — 192.168.1.0/24

Device	Interface	IP Address	Subnet Mask	Gateway
Router1	Gig0/0	192.168.1.1	255.255.255.0	—
PC1	NIC	192.168.1.10	255.255.255.0	192.168.1.1
PC2	NIC	192.168.1.11	255.255.255.0	192.168.1.1
PC3	NIC	192.168.1.12	255.255.255.0	192.168.1.1

3.2 WAN Link — 10.0.0.0/30

Device	Interface	IP Address	Subnet Mask
Router1	Gig0/1	10.0.0.1	255.255.255.252
Router2	Gig0/0	10.0.0.2	255.255.255.252

3.3 Network 2 — 192.168.2.0/24

Device	Interface	IP Address	Subnet Mask	Gateway
Router2	Gig0/1	192.168.2.1	255.255.255.0	—
PC4	NIC	192.168.2.10	255.255.255.0	192.168.2.1
PC5	NIC	192.168.2.11	255.255.255.0	192.168.2.1
PC6	NIC	192.168.2.12	255.255.255.0	192.168.2.1
PC7	NIC	192.168.2.13	255.255.255.0	192.168.2.1

4. Router Configuration

4.1 Router 1 CLI Commands

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface gigabitEthernet 0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface gigabitEthernet 0/1
R1(config-if)#ip address 10.0.0.1 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
R1#write memory
```

4.2 Router 2 CLI Commands

```
R2(config)#interface gigabitEthernet 0/0
R2(config-if)#ip address 10.0.0.2 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface gigabitEthernet 0/1
R2(config-if)#ip address 192.168.2.1 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
R2#
```

5. Connectivity Test Results

Ping tests were performed from PC1 to all devices across both networks to verify connectivity.

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=11ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 3ms
```

```
C:\>ping 192.168.2.11

Pinging 192.168.2.11 with 32 bytes of data:

Reply from 192.168.2.11: bytes=32 time<1ms TTL=126
Reply from 192.168.2.11: bytes=32 time<1ms TTL=126
Reply from 192.168.2.11: bytes=32 time=10ms TTL=126
Reply from 192.168.2.11: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

```
C:\>ping 192.168.2.12

Pinging 192.168.2.12 with 32 bytes of data:

Reply from 192.168.2.12: bytes=32 time=10ms TTL=126
Reply from 192.168.2.12: bytes=32 time=10ms TTL=126
Reply from 192.168.2.12: bytes=32 time<1ms TTL=126
Reply from 192.168.2.12: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 5ms
```

```
C:\>ping 192.168.2.13

Pinging 192.168.2.13 with 32 bytes of data:

Reply from 192.168.2.13: bytes=32 time<1ms TTL=126
Reply from 192.168.2.13: bytes=32 time=11ms TTL=126
Reply from 192.168.2.13: bytes=32 time<1ms TTL=126
Reply from 192.168.2.13: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 2ms
```

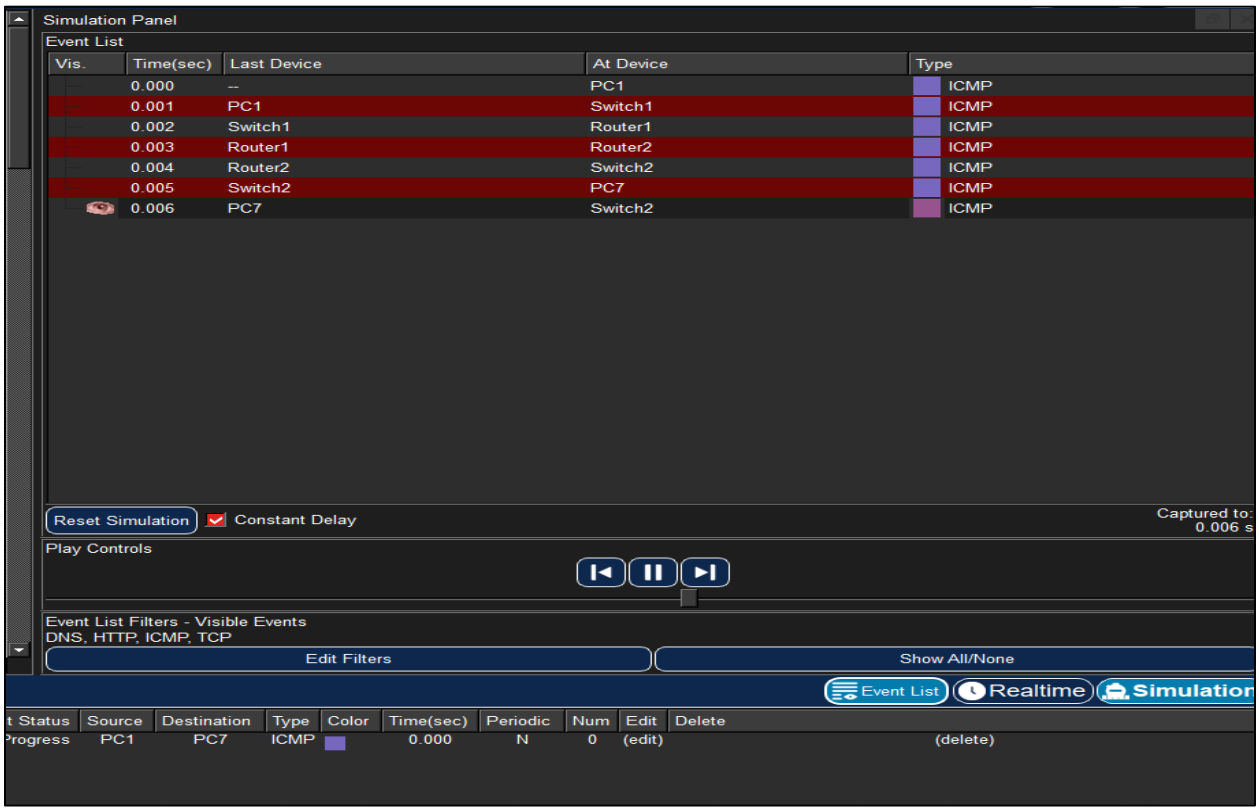
Result: All ping tests were successful with 0% packet loss confirming full connectivity across both networks.

6. Traffic Analysis — Simulation Mode

Cisco Packet Tracer Simulation Mode was used to capture and analyze network traffic at each device hop.

6.1 PDU Packet Analysis (PC1 to PC7)

Hop	Device	Layer 2 (MAC)	Layer 3 (IP)	TTL
1	PC1	Source MAC generated	192.168.1.10	128
2	Switch1	MAC forwarded	192.168.1.10	128
3	Router1	MAC address changed	192.168.1.10	127
4	Router2	MAC address changed	192.168.1.10	126
5	Switch2	MAC forwarded	192.168.2.13	126
6	PC7	Destination MAC matched	192.168.2.13	126



6.2 Key Observations

- Source IP (192.168.1.10) and Destination IP (192.168.2.13) remained the same throughout all hops
- MAC address changed at each Router — confirming Layer 2 rewriting
- TTL decreased by 1 at each Router hop (128 → 127 → 126)
- ICMP Type 0x08 confirms the traffic was a Ping (Echo Request)

7. Performance Analysis — Traceroute

Traceroute was executed from PC1 to PC7 (192.168.2.13) to identify the path and measure latency at each hop.

7.1 Traceroute Results

Hop	IP Address	Latency	Device / Meaning
1	192.168.1.1	1 ms	Router1 — Local Gateway (LAN)
2	10.0.0.2	0 ms	Router2 — WAN Link
3	192.168.2.13	6 ms	PC7 — Destination Reached

```
C:\>tracert 192.168.2.13

Tracing route to 192.168.2.13 over a maximum of 30 hops:

  1   0 ms      1 ms      1 ms      192.168.1.1
  2   0 ms      0 ms      0 ms      10.0.0.2
  3   0 ms      6 ms      0 ms      192.168.2.13

Trace complete.
```

7.2 Bottleneck Analysis

- Total Hops: 3 (PC1 → Router1 → Router2 → PC7)
- Bottleneck Location: Hop 3 showed the highest latency (6ms) at the destination
- The WAN link between Router1 and Router2 is the potential bottleneck in a real network
- Trace Status: Complete — all hops responded successfully

8. Fault Simulation and Detection

Three network faults were deliberately introduced and detected to demonstrate fault monitoring capabilities.

8.1 Fault 1 — Router Interface Shutdown

```
C:\>ping 192.168.2.13

Pinging 192.168.2.13 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 192.168.2.13:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

8.2 Fault 3 — Wrong IP Address

```
C:\>ping 192.168.2.14

Pinging 192.168.2.14 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.14:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

8.4 Fault Summary

Fault	Cause	Symptom	Resolution
Fault 1	Interface shutdown	Host unreachable	no shutdown
Fault 2	Wrong IP address	Request timed out	Correct IP

9. Conclusion and Recommendations

9.1 Conclusion

This project successfully demonstrated network monitoring and performance analysis using Cisco Packet Tracer. The following was achieved:

- A two-network topology was built and fully configured with static routing
- All 7 PCs communicated successfully across both networks with 0% packet loss
- Network traffic was captured and analyzed using Simulation Mode — observing TTL, MAC, and IP behavior at each hop
- Traceroute identified the packet path and measured latency at each hop
- Three network faults were simulated, detected, and successfully resolved

9.2 Recommendations

- Use redundant WAN links to eliminate single point of failure between routers
- Implement dynamic routing protocols (OSPF or EIGRP) instead of static routes for larger networks
- Add a DNS server and HTTP server for more comprehensive traffic analysis
- Use VLANs to segment network traffic for better performance and security
- Deploy network monitoring tools like SNMP for real-time fault detection in production environments